

Data Warehouse - 20 Toughest MCQs (Correct Answers Marked)

1. Which of the following is NOT a characteristic of a Data Warehouse?

- A) Subject-Oriented
- B) Non-Volatile
- C) Real-Time Processing ✓**
- D) Integrated

2. Which component of a Data Warehouse is responsible for managing metadata?

- A) ETL Process
- B) OLAP Engine
- C) Metadata Repository ✓**
- D) Staging Area

3. Which of the following is a major drawback of an OLTP system in the context of analytical processing?

- A) It lacks historical data storage ✓**
- B) It cannot support large numbers of transactions
- C) It supports batch processing instead of real-time updates
- D) It does not support normalization

4. Which type of OLAP stores pre-aggregated data for faster query performance?

- A) MOLAP ✓**
- B) ROLAP
- C) HOLAP
- D) DOLAP

5. In Dimensional Modeling, what does a slowly changing dimension (SCD) handle?

- A) Historical changes in data ✓**
- B) Real-time data processing
- C) Data aggregation
- D) ETL failures

6. What is the key advantage of a Data Lake compared to a Data Warehouse?

- A) Structured storage only
- B) Schema-on-read approach ✓**
- C) High cost and complexity
- D) Limited scalability

7. Which of the following is NOT a component of a Data Warehouse architecture?

- A) Data Source
- B) OLAP Server
- C) Data Mover
- D) Metadata Repository ✗ (Correct: C)**

8. What is the main difference between a Data Warehouse and a Data Mart?

- A) A Data Mart is larger than a Data Warehouse
- B) A Data Warehouse is subject-oriented, while a Data Mart is application-specific ✓**

- C) A Data Mart contains normalized data
- D) A Data Warehouse does not support aggregation

9. The Star Schema design has which key characteristic?

- A) Highly normalized structure
- B) Uses fact tables with multiple relationships
- C) Single fact table surrounded by denormalized dimension tables ✓**
- D) Uses only primary keys for relationships

10. Which of the following is a disadvantage of the Snowflake Schema?

- A) It does not support normalization
- B) It increases redundancy
- C) It leads to slower query performance due to complex joins ✓**
- D) It requires denormalization of data

11. What is a key benefit of using an OLAP system over an OLTP system?

- A) Faster transactional updates
- B) Supports real-time analytics
- C) Pre-aggregated data for faster query performance ✓**
- D) Higher data normalization

12. What is the role of ETL in a Data Warehouse?

- A) Only loads data into the warehouse
- B) Extracts, Transforms, and Loads data from various sources ✓**
- C) Performs only data cleaning and validation
- D) Handles direct OLAP queries

13. What differentiates a Fact Table from a Dimension Table?

- A) Fact Tables store detailed transactional data, while Dimension Tables store descriptive attributes ✓**
- B) Fact Tables contain categorical information
- C) Dimension Tables store numerical data
- D) Fact Tables are not used in OLAP

14. What is a surrogate key in a Data Warehouse?

- A) A business-defined primary key
- B) A foreign key referencing a fact table
- C) A system-generated unique identifier ✓**
- D) A natural key from the source system

15. Why is data denormalization often used in Data Warehousing?

- A) To reduce redundancy
- B) To optimize OLAP query performance ✓**
- C) To enforce referential integrity
- D) To minimize data storage requirements

16. Which of the following is NOT an ETL tool?

- A) Informatica PowerCenter
- B) Talend

C) SAP BW

D) MongoDB ✓

17. In Slowly Changing Dimension Type 2 (SCD Type 2), how are historical changes handled?

A) Overwriting the old record

B) Creating a new row with a surrogate key ✓

C) Ignoring the change

D) Updating the primary key

18. Which of the following is an example of a degenerate dimension?

A) Customer Name

B) Order ID stored in a Fact Table ✓

C) Product Category

D) Date Dimension

19. What is the primary advantage of a Columnar Database for Data Warehousing?

A) Better support for transactional processing

B) Optimized for analytical queries and compression ✓

C) Stores only unstructured data

D) Eliminates the need for indexing

20. Why is a conformed dimension important in a Data Warehouse?

A) It is only used in OLTP systems

B) It ensures consistency across multiple fact tables ✓

C) It does not support historical tracking

D) It is an alternative to surrogate keys

1. In Informatica, which component is responsible for managing metadata and security?

A) PowerCenter Repository ✓

B) PowerCenter Designer

C) PowerCenter Workflow Manager

D) PowerCenter Workflow Monitor

2. What is the primary function of an Expression Transformation in Informatica?

A) Filtering rows

B) Performing row-wise calculations ✗ (Your answer: C, Correct: B)

C) Joining multiple sources

D) Aggregating records

3. Which of the following is TRUE about Router Transformation?

A) It filters data based on a single condition

B) It acts like multiple Filter transformations ✓

C) It is used only in dynamic lookups

D) It cannot have multiple output groups

4. What is the primary purpose of a Filter Transformation?

A) To distribute records to multiple groups

B) To filter out unwanted records before further processing ✓

- C) To perform calculations on individual rows
- D) To combine multiple source tables

5. In Aggregator Transformation, what is the purpose of the 'Sorted Input' option?

- A) To allow duplicate records
- B) To improve performance by grouping sorted data** ✓
- C) To enable real-time processing
- D) To remove duplicate values automatically

6. Which join type is NOT supported by the Joiner Transformation in Informatica?

- A) Inner Join
- B) Full Outer Join
- C) Left Outer Join
- D) Cross Join** ✓

7. What is the primary function of a Rank Transformation?

- A) Sorting records in descending order
- B) Assigning a rank to each row based on a specific column** ✓
- C) Eliminating duplicate records
- D) Performing mathematical calculations

8. How does Sequence Generator Transformation help in Informatica?

- A) Generates unique numbers sequentially** ✓
- B) Generates random numbers
- C) Sorts records based on a sequence
- D) Creates primary keys for relational tables

9. What is the key difference between Connected and Unconnected Lookup?

- A) Unconnected Lookup returns multiple rows, while Connected Lookup returns only one
- B) Connected Lookup is faster than Unconnected Lookup
- C) Connected Lookup receives input directly from the pipeline, while Unconnected Lookup is invoked using an expression** ✗ (Your answer: A, Correct: C)
- D) Unconnected Lookup cannot be used in transformations

10. Which of the following is NOT an option in Slowly Changing Dimensions (SCD) in Informatica?

- A) Type 1 – Overwrite** ✓
- B) Type 2 – Versioning
- C) Type 3 – Add New Column
- D) Type 4 – Archive in a different database

11. How can you optimize a Lookup Transformation in Informatica?

- A) By using a dynamic lookup cache** ✓
- B) By enabling sorted input
- C) By increasing the session timeout
- D) By using a full outer join

12. What does the Update Strategy Transformation allow you to do?

A) Only insert new records into the target

B) Define row-level operations like Insert, Update, Delete, or Reject ✓

C) Perform bulk updates only

D) Convert data types dynamically

13. What is the purpose of a Dynamic Lookup Cache?

A) It refreshes lookup data at the start of every session

B) It allows lookup cache updates when new data is found ✓

C) It prevents updates to lookup values

D) It only works with SQL override

14. When would you use a Union Transformation in Informatica?

A) To join two tables based on a key

B) To combine data from multiple pipelines with different structures ✓

C) To remove duplicate records

D) To perform a Cartesian join

15. In Informatica, how does the CDC (Change Data Capture) process work?

A) Captures only inserted records

B) Captures only updated records

C) Captures Insert, Update, and Delete operations from source systems ✓

D) Only works with flat files

16. What is a key benefit of using Pushdown Optimization in Informatica?

A) Moves transformation logic to the database for performance improvement ✓

B) Pushes data into multiple staging areas before loading

C) Increases ETL session execution time

D) Enables session logs to be stored in the repository

17. What happens if a session in Informatica fails due to a transformation error?

A) The session automatically restarts

B) The failed records are logged in the bad file and session fails ✓

C) The error is ignored, and the session continues

D) The workflow continues with no error logging

18. Which of the following statements about Parameter Files in Informatica is TRUE?

A) They contain only database connection details

B) They can be modified only at runtime

C) They store values that can be dynamically assigned during workflow execution ✓

D) They cannot be used in mapping variables

19. What is the main advantage of using Partitioning in Informatica?

A) Reduces data redundancy in the target

B) Improves session performance by parallel processing ✓

C) Automatically removes NULL values

D) Prevents duplicate records from entering the target

20. In Informatica, what is the function of the Session Log file?

- A) Stores metadata for transformations
- B) Logs errors, statistics, and session execution details** ✓
- C) Stores processed data before loading into the target
- D) Only records warnings, not errors

Tableau - 20 Toughest MCQs (Correct Answers Marked in Bold)

1. In Tableau, what is the primary function of a Dimension?

- A) To store numeric values for aggregation
- B) To define categories and qualitative data** ✓
- C) To filter data in worksheets
- D) To perform mathematical operations

2. What happens when two tables are joined in Tableau without a common key?

- A) The join fails
- B) A Cartesian join occurs, multiplying the number of rows** ✓
- C) A null value is returned
- D) Tableau automatically creates a common key

3. Which of the following chart types is best for comparing trends over time?

- A) Scatter Plot
- B) Pie Chart
- C) Line Chart** ✓
- D) Histogram

4. What is the default join type in Tableau when blending data sources?

- A) Inner Join** ✓
- B) Left Join
- C) Right Join
- D) Full Outer Join

5. What is a 'LOD Expression' in Tableau used for?

- A) To improve dashboard loading speed
- B) To perform calculations at different levels of granularity** ✓
- C) To connect multiple data sources
- D) To create dynamic filters

6. Which of the following best describes a calculated field in Tableau?

- A) A dynamically updated field based on a formula** ✓
- B) A pre-defined field stored in the dataset
- C) A field that can only be used in filters
- D) A field that is only available in Data Source mode

7. In Tableau, what is the difference between a Live Connection and an Extract?

- A) A Live Connection updates in real-time, while an Extract is a static snapshot of data** ✓
- B) An Extract always runs faster than a Live Connection
- C) A Live Connection requires data blending, while an Extract does not
- D) Extracts do not support calculated fields

8. What is the primary purpose of the 'Pages' shelf in Tableau?

- A) To create animated visualizations based on a dimension** ✓
- B) To filter data dynamically
- C) To group similar data points
- D) To create hierarchical relationships

9. What does a 'Data Extract' in Tableau do?

- A) Refreshes the entire database every time you open the workbook
- B) Creates a static snapshot of data for faster performance** ✓
- C) Deletes unnecessary columns from the dataset
- D) Converts data into JSON format

10. Which function is used to create table calculations in Tableau?

- A) SUM()
- B) INDEX()
- C) WINDOW_SUM()
- D) Both B & C** ✓

11. In Tableau, how do you combine two different datasets that do not share a common key?

- A) By using Data Blending** ✓
- B) By creating a Relationship
- C) By using an Inner Join
- D) By adding a cross-database join

12. What is the difference between 'Continuous' and 'Discrete' fields in Tableau?

- A) Continuous fields are used for text data, while Discrete fields are used for numbers
- B) Continuous fields create axes, while Discrete fields create headers** ✓
- C) Continuous fields are always measures, while Discrete fields are always dimensions
- D) Discrete fields cannot be used in calculations

13. What does the 'Fixed' keyword do in a Level of Detail (LOD) expression?

- A) It creates a fixed reference line in a chart
- B) It calculates an aggregate value at a specified level of granularity** ✓
- C) It locks a filter so that it cannot be changed
- D) It prevents data from updating dynamically

14. Which type of Tableau filter is applied before data is loaded into the view?

- A) Extract Filter
- B) Dimension Filter
- C) Context Filter** ✓
- D) Measure Filter

15. What is the purpose of the 'Measure Names' and 'Measure Values' fields in Tableau?

- A) To allow multiple measures to be displayed in a single visualization** ✓
- B) To create calculated fields dynamically
- C) To define relationships between dimensions
- D) To control user-level security

16. What happens when you use the 'Dual Axis' feature in Tableau?

A) Two measures share the same axis

B) Two measures are plotted on separate synchronized axes ✗ (Your answer: A, Correct: B)

C) A scatter plot is automatically created

D) A hierarchical relationship is formed

17. How does Tableau handle NULL values in a dataset?

A) It automatically removes them

B) It displays them as blanks in visualizations

C) It replaces them with zero ✗ (Your answer: C, Correct: B)

D) It creates an error in the calculation

18. In Tableau, which of the following is TRUE about Context Filters?

A) They are applied before all other filters

B) They improve performance by reducing data size before applying other filters ✓

C) They cannot be used with Measure filters

D) They work only with extracts, not live connections

19. What is the purpose of 'Tableau Extensions'?

A) To connect Tableau with third-party applications and add custom functionality ✓

B) To create advanced table calculations

C) To perform statistical analysis within Tableau

D) To enable cloud storage integration

20. What is the difference between a Dashboard and a Story in Tableau?

A) A Dashboard contains multiple sheets, while a Story is a sequence of visualizations ✓

B) A Story is interactive, while a Dashboard is static

C) A Dashboard is only used for filtering data, while a Story is used for presentations

D) A Story only works with a single data source